

Scientific Computing WS25/26

Working With Data

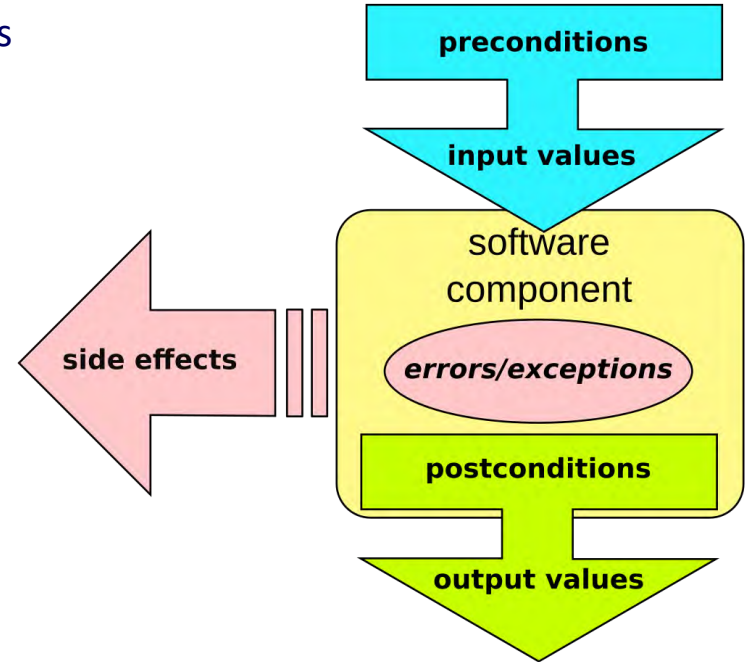
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Design by Contract (DbC)

- Similar to how business contracts are determined between clients and suppliers, software designers define precise and verifiable specifications for software components to ensure correctness.
- When designing a contract, the designer should always consider and answer these three questions:
 1. What does the contract expect?
 2. What does the contract guarantee?
 3. What does the contract maintain?

- The contract defines the following pieces of information required for an operation between the client components and server component:
 - Preconditions
 - Postconditions
 - Invariants
 - (Un)acceptable input values or types
 - Return values or types
 - Error condition values or types that can occur
 - Side effects
 - Performance guarantees



- While not a *replacement* for testing, DbC complements the process by implementing self-tests that detect errors before they transform into invalid results
- With contracts in place, suppliers don't have to verify the contract conditions are satisfied (i.e. offensive programming)
- DbC also facilitates code reuse as the contract for each piece of code is fully documented

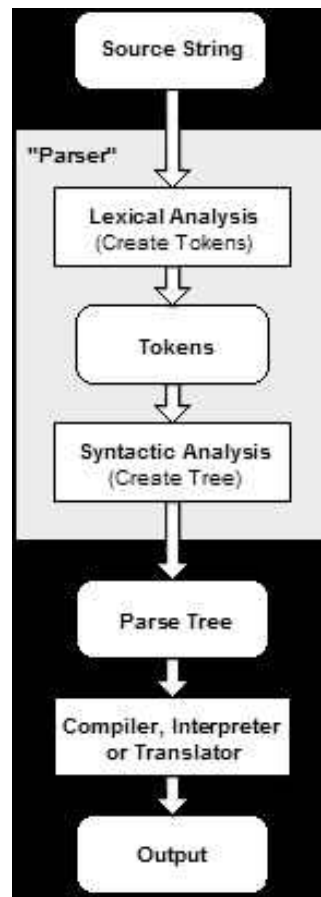
CFG and Parsing

- EBNF is a code that expresses a context-free grammar (the syntax of a language)
- Often represented as a family of notations consisting of terminals and non-terminal production rules which restrict how terminals can be combined into a valid sequence

```
digit excluding zero = "1" | "2" | "3" | "4" | "5" | "6" | "7" | "8" | "9" ;  
digit                = "0" | digit excluding zero ;
```

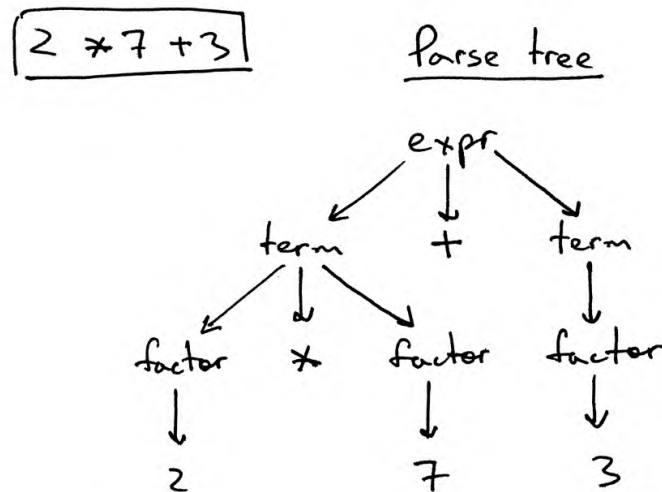
```
positive integer = digit excluding zero, { digit } ;
```

- A process of analyzing a string of symbols in a language or data structure, and breaking it into parts conforming to the rules of a formal grammar
1. Analyzer to deconstruct input into tokens
 2. Tokens added to Parse Tree
 3. If expression fits grammar, take the appropriate action (e.g. compute expression, generate code)



Parse trees are an effective way to analyze that tokens follow the correct syntax and form an allowable expression:

- The tokens here would be 2, *, 7, +, 3
- Analyzer enforces rules that * and + mark the start of a new token
- Expression is allowable so can proceed to evaluation



A top-down parser built from a set of mutually recursive procedures where each procedure implements one of the non-terminals of the grammar.

Example grammar in LL(1) form:

```
expression = ["+" | "-"] term {"+" | "-"} term .
```

```
term = factor {"*" | "/" } factor .
```

```
factor =  
    ident  
    | number  
    | "(" expression ")" .
```

C Implementation of RDP:

```
void factor(void) {  
    if (accept(ident)) {  
        ;  
    } else if (accept(number)) {  
        ;  
    } else if (accept(lparen)) {  
        expression();  
        expect(rparen);  
    } else {  
        error("factor: syntax error");  
        nextsym();  
    }  
}  
  
void term(void) {  
    factor();  
    while (sym == times || sym == slash) {  
        nextsym();  
        factor();  
    }  
}  
  
void expression(void) {  
    if (sym == plus || sym == minus)  
        nextsym();  
    term();  
    while (sym == plus || sym == minus) {  
        nextsym();  
        term();  
    }  
}
```

Unicode

- Unicode is a universal character set that makes it possible to design code to handle the requirements of all languages and regions at the same time
- All modern operating systems, computing environments, programming languages, and applications support the core of the Unicode Standard.
- The Unicode Consortium is a non-profit organization that coordinates Unicode's development



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U+09F0	U+12E4	U+516B	U+01E7	U+0A68
𑌨	◡	🚩	¿	ॐ
U+178A	U+1D55	U+2690	U+00BF	U+0994

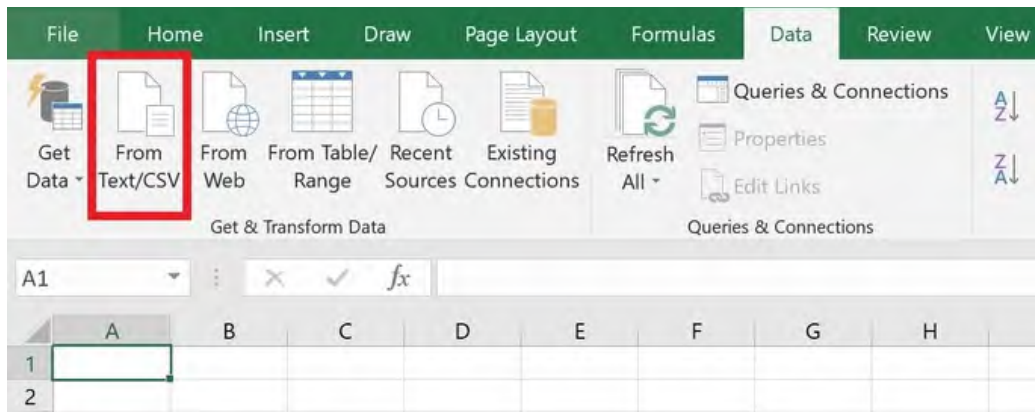
- Can encode up to roughly 1.1 million characters and 3,790 emojis
- Encoding Formats (UTF-8, UTF-16, etc.)

- Represented as U+XXXX
- U+0041 is the Latin capital letter “A”
- U+1F643 is the upside-down smiley emoji 😞

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U+3063	U+06F6	U+3142	U+1F495	U+2460
ㄣ	꺄	..	😓	⊖
U+0F3D	U+CB04	U+2025	U+1F605	U+0398

Data Formats

- Comma separated values format (CSV) as a way to store data is widely supported by a variety of software and is commonly used in many data and spreadsheet applications
- Used for interchanging database information between machines of different architectures to avoid incompatibilities such as byte-order and word size
- Benefits include the simple formatting, human readability, and less data inaccuracies than traditional fixed-column-aligned formats



PurchasedItems.csv - Notepad

File Edit Format View Help

```
Date,Weekday,Region,Employee,Item, Units , Unit Cost , Total
15-Dec-21,Wednesday,Central,Jones,Pen Set,700, $1.99 , " $1,393.00 "
16-Dec-21,Thursday,West,Kivell,Binder,85, $19.99 , " $1,699.15 "
17-Dec-21,Friday,Central,Howard,Pen & Pencil,62, $4.99 , $309.38
18-Dec-21,Saturday,East,Gill,Pen,58, $19.99 , " $1,159.42 "
19-Dec-21,Sunday,East,Anderson,Binder,10, $4.99 , $49.90
20-Dec-21,Monday,East,Anderson,Pen Set,19, $2.99 , $56.81
21-Dec-21,Tuesday,East,Anderson,Pen Set,6, $1.99 , $11.94
22-Dec-21,Wednesday,Central,Howard,Pen & Pencil,10, $4.99 , $49.90
23-Dec-21,Thursday,West,Wilson,Paper,39, $1.99 , $77.61
24-Dec-21,Friday,West,Wilson,Binder,1, $8.99 , $8.99
25-Dec-21,Saturday,West,Wilson,Pen & Pencil,80, $4.99 , $399.20
26-Dec-21,Sunday,West,Wilson,Binder,51, $1.99 , $101.49
27-Dec-21,Monday,West,Wilson,Binder,10, $19.99 , $199.90
28-Dec-21,Tuesday,West,Wilson,Pen Set,15, $4.99 , $74.85
29-Dec-21,Wednesday,West,Wilson,Desk,31, $125.00 , " $3,875.00 "
30-Dec-21,Thursday,Central,Jones,Pen Set,46, $15.99 , $735.54
31-Dec-21,Friday,West,Kivell,Binder,61, $8.99 , $548.39
1-Jan-22,Saturday,Central,Jones,Pen,90, $8.99 , $809.10
```

Ln 1, Col 1 100% Windows (CRLF) UTF-8

- Extensible Markup Language (XML) is a language and file format for storing, transmitting, and reconstructing data
- Defines a set of rules for encoding documents in a format that is both human-readable and machine-readable
- The design goals of XML emphasize simplicity, generality, and usability across the internet
- XML documents consist entirely of characters from the Unicode repertoire

```
<?xml version="1.0" encoding="UTF-8" standalone="yes" ?>
- <companies>
- <company>
  <companyname>Stanford and
    Son</companyname>
  - <employee>
    <code>1</code>
    <name>Joe Jackson</name>
    <street>14th street</street>
    <houseno>1</houseno>
    <areacode>1050 DD</areacode>
    <place>NoWhere</place>
    <phone>0100 987654</phone>
  </employee>
  - <employee>
    <code>2</code>
    <name>Peter de Wit</name>
    <street>ChurchLane</street>
    <houseno>4a</houseno>
    <areacode>9876 AB</areacode>
    <place>Whereever</place>
    <phone>0100 987654</phone>
  </employee>
  - <employee>
    <code>3</code>
    <name>John Brown</name>
    <street>1st street</street>
    <houseno>243</houseno>
    <areacode>5558 ZZ</areacode>
    <place>OutSide</place>
    <phone>0333 999888</phone>
  </employee>
</company>
</companies>
```

- APIs for XML processing tend to fall into the categories of **SAX** (stream-oriented) or **DOM** (tree-traversal):

DOM

- Document Object Model (DOM) treats an XML document as a logical tree structure wherein each node is an object representing a part of the document.
- By allowing programmatic access to the tree, one can change the structure, style or content of a document
- Memory intensive

SAX

- Simple API for XML (SAX) is an event-driven algorithm for parsing XML documents.
- Instead of building a syntax tree, SAX parsers make a single pass through the document, operating on each piece sequentially and issuing parsing events
- Memory efficient



- JavaScript Object Notation (JSON) is a commonly used open standard file format that uses human-readable text to store and transmit data objects consisting of name–value pairs and arrays.
- Grew out of a need to replace outdated browser plugins like Flash in the early 2000s
- Diverse uses in web applications by providing an abstraction layer for developers to create stateful applications and hold HTTP connections with servers
- Supports the full Unicode character set
- Some relational databases have added support for native JSON data types

```
{
  "first_name": "John",
  "last_name": "Smith",
  "is_alive": true,
  "age": 27,
  "address": {
    "street_address": "21 2nd Street",
    "city": "New York",
    "state": "NY",
    "postal_code": "10021-3100"
  },
  "phone_numbers": [
    {
      "type": "home",
      "number": "212 555-1234"
    },
    {
      "type": "office",
      "number": "646 555-4567"
    }
  ],
  "children": [
    "Catherine",
    "Thomas",
    "Trevor"
  ],
  "spouse": null
}
```

- Lightweight and efficient but less suitable for highly complex data
- Common data types: Number, String, Boolean, Array, Object
- Order does not matter
- No support for comments

Both formats support data interchanging and have widespread support for creation, reading, and decoding in real-world situations but have some key differences:

XML

- Separates data from metadata
- Each attribute has a single value and represented with tags `</>`
- Address individual nodes with unique ID
- Values are only string characters
- Permits strong typing and formal structure
- Supports comments

JSON

- Low overhead alternative to XML
- Attributes represented with curly braces `{ }` and double quotes
- Key-to-value mapping for objects
- JSON keys can contain special characters
- Does not support comments

Endianness is the order in which bytes are stored or transmitted for multi-byte data.

- Big-endian (BE) system stores the most significant byte at the smallest memory address and the least significant byte at the largest
- Little-endian (LE) system stores the least-significant byte at the smallest address.

